

FACTORIAL-RESIDUE HYPERGRAPHS AND COVERING CONGRUENCES

ABSTRACT. We study covering congruences for factorials using a weighted hypergraph whose edges collect indices with equal factorial residues and whose colors are primes. The minimum weight $\kappa(L)$ of a rainbow cover equals the logarithm of the least squarefree modulus of such a system. We prove $\kappa(L) \leq (1 - \delta)L \log L$ for an absolute $\delta > 0$, even after any $\lfloor L/100 \rfloor$ primes are excluded. By counting permutations of the remaining assignments, we obtain one modulus with $((L + 1)!)^{1/2 - o(1)}$ distinct covering classes. This gives at least as many L -rough integers in $(2L!, (L + 1)!]$ whose differences from $1!, \dots, L!$ are all composite, uniformly in any prescribed reduced residue class modulo $d \leq e^L$. We also prove $\kappa(L) \geq (2/3 - o(1))L^{1/3} \log L$. A resultant estimate bounds the number of triples of equal factorial residues with small span. It implies that covers whose consecutive triples all have span $o(L^{1/4})$ have weight at least $(1/2 - o(1))L \log L$. The rough-integer result strengthens a quantitative solution of an auxiliary problem of Erdős; the integers obtained need not be prime.

Working draft — exposition and mathematical review in progress.

1. INTRODUCTION

In Problem A2 of [4], Erdős asks whether there are infinitely many primes p such that $p - k!$ is composite whenever $1 \leq k! < p$. This is also listed as Erdős Problem 1059 [2]. He also suggested an auxiliary problem in which the integer itself need only have large prime factors. In this paper, we study the covering congruences used to construct such integers. When several factorials have the same residue modulo a prime, one congruence makes that prime divide each of the corresponding differences. We use these equalities to reduce the product of the primes needed to cover all the factorials.

We first fix the notation for this covering problem. For $L \geq 2$, let $\mathcal{H}_L$ be the weighted edge-colored multihypergraph on $V_L = \{2, \dots, L\}$ whose nonempty edges are

$$E_{q,a} = \{k \in V_L : k! \equiv a \pmod{q}\}, \quad L < q < L!, \quad a \in (\mathbb{Z}/q\mathbb{Z})^\times.$$

Each edge has color q and weight $\log q$. A *rainbow cover* selects at most one edge of each color and covers every vertex. Write $\kappa(L)$ for its minimum weight, with value $+\infty$ when no cover exists. The Chinese remainder theorem

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(CRT) gives

$$\kappa(L) = \log M_{\min}(L),$$

where $M_{\min}(L)$ is the least squarefree modulus of a compatible covering system using these primes and residues. We give the precise definition and prove this equality in Proposition 2.5. Our first result bounds the weight of a cover.

Theorem 1.1. *There are absolute constants $\delta > 0$ and L_0 such that*

$$\kappa(L) \leq (1 - \delta)L \log L$$

for every integer $L \geq L_0$.

Assigning a different prime to each index gives a cover of weight $(1 + o(1))L \log L$. Theorem 1.1 improves this bound by pairing indices. The identity used in the proof is

$$(1) \quad (i + 2)! - i! = i!(i^2 + 3i + 1).$$

A product argument and a dimension-one upper-bound sieve give a positive proportion of pairs with primes $L < q \ll L^{2-\alpha}$. We then use a conflict graph to choose disjoint pairs with distinct primes. Each chosen prime covers both indices in its pair. We assign individual primes of size $O(L \log L)$ to the remaining indices.

The same construction allows us to exclude any set of at most $\lfloor L/100 \rfloor$ primes; see Theorem 5.2. This will be useful when we prescribe an additional congruence for the integers we construct. We also count the distinct residue classes obtained by permuting the primes assigned to individual indices in Theorem 6.5.

We next consider lower bounds. Proposition 6.1 and Corollary 6.2 give

$$\kappa(L) \geq (2/3 - o(1))L^{1/3} \log L.$$

There is a stronger bound when the indices in each fiber are close together. Suppose every consecutive triple in every designated fiber has span at most $cL^{1/4}$, where $c > 0$ is fixed. Then Corollary 7.5 gives

$$\frac{w(\mathcal{C})}{\log((L+1)!)} \geq \frac{1}{2} - \frac{c^4}{192} - o(1).$$

In particular, a full cover whose consecutive-triple spans are $o(L^{1/4})$ costs at least $(1/2 - o(1))L \log L$. The order of $\kappa(L)$ is not determined by these bounds.

Erdős's auxiliary problem. The auxiliary problem of Erdős asks for infinitely many pairs (L, n) with $L! < n \leq (L+1)!$, every prime factor of n exceeding L , and $n - k!$ composite for $1 \leq k \leq L$. For $n > 1$, write $P^-(n)$ and $P^+(n)$ for its least and greatest prime factors. We call n *L-rough* when $P^-(n) > L$.

The covering construction gives the following quantitative form of the auxiliary statement.

Theorem 1.2. *There are absolute constants $\eta > 0$ and L_1 such that, for every integer $L \geq L_1$, at least*

$$\frac{((L+1)!)^\eta}{100 \log L}$$

integers $n \in (2L!, (L+1)!]$ are L -rough and have $n - k!$ composite for all $1 \leq k \leq L$.

By counting several covering classes, we obtain the following stronger bound, also in prescribed arithmetic progressions.

Theorem 1.3. *For every $\varepsilon > 0$, there is L_ε such that the following holds for every integer $L \geq L_\varepsilon$. Put $X = (L+1)!$. For every integer $1 \leq d \leq e^L$ and every residue $b \pmod{d}$ with $(b, d) = 1$, at least $X^{1/2-\varepsilon}$ integers $n \in (2L!, X]$ satisfy*

$$n \equiv b \pmod{d}, \quad P^-(n) > L, \quad n - k! \text{ composite } (1 \leq k \leq L).$$

The threshold is independent of d and b .

Our proof of Theorem 1.3 uses the elementary estimate

$$\#\{(i, j) \in \{1, \dots, L\}^2 : i! \equiv j! \pmod{q}\} \leq 6L^{3/2} \quad (q > L),$$

and Lemma 6.4, which bounds the number of distinct images of permutations. Estimates of order $L^{3/2}$ for equal factorial residues are classical; see [3, Theorem 1, with $\ell = 1$]. A direct proof with the displayed constant is included. We then count rough integers in these classes by inclusion–exclusion.

Land [8] has posted a proposed solution of the prime problem using a sieve for sparse sets of shifts and Fourier estimates. The theorem stated there also implies existence in the auxiliary problem. Our proofs are independent of that work. Here we study the size of the covering modulus, the number of classes it gives, and restrictions on the indices that one prime can cover.

Other work on factorial residues studies value sets and congruence counts modulo a fixed prime; see [1, 3, 6, 5]. In our covering problem, the prime varies from one edge to another, but only one residue can be chosen for each prime.

Outline of the paper. Section 2 defines the covers and proves the elementary bounds used later. Sections 3–5 give the quadratic construction, including the choice of primes when some are excluded. Section 6 counts covering classes, and Section 7 bounds triples of equal factorial residues with small span. In Section 8, we count rough integers in the resulting progressions and prove Theorems 1.2 and 1.3. We end with computational examples and further questions.

The original quantitative auxiliary theorem and its proof using quadratic congruences have an accompanying Lean formalization registered as PALOMAR-2026-08-27-000003 v1 [9]. The standard dimension-one upper-bound sieve is an explicit external hypothesis, matched to [7, Theorem 18.11(a)]. That formalization does not include the additional results

here on excluded primes, counts of covering classes, resultants, or prescribed progressions.

2. RESIDUE COVERS AND COVERING CONGRUENCES

Let $L \geq 2$ and put

$$V_L = \{2, 3, \dots, L\}.$$

Definition 2.1. For a prime $L < q < L!$ and a nonzero residue $a \pmod{q}$, let $e_{q,a}$ be a labeled hyperedge with incident vertex set

$$E_{q,a} = \{k \in V_L : k! \equiv a \pmod{q}\}.$$

After empty edges are omitted, these labeled edges form a weighted edge-colored multihypergraph $\mathcal{H}_L$. The edge $e_{q,a}$ has color q and weight $\log q$.

Since $q > L$, every $k!$ with $k \in V_L$ is nonzero modulo q . Therefore, for each fixed q , the fibers $E_{q,a}$ partition V_L , with empty fibers omitted.

Definition 2.2. A *rainbow edge cover* of $\mathcal{H}_L$ is a finite collection

$$\mathcal{C} = \{e_{q,a_q} : q \in Q\}$$

with distinct primes $L < q < L!$, one residue $a_q \in (\mathbb{Z}/q\mathbb{Z})^\times$ for each $q \in Q$, nonempty fibers E_{q,a_q} , and

$$V_L \subseteq \bigcup_{q \in Q} E_{q,a_q}.$$

Its weight is

$$w(\mathcal{C}) = \sum_{q \in Q} \log q.$$

We define

$$\kappa(L) = \inf_{\mathcal{C}} w(\mathcal{C}),$$

where the infimum is over all rainbow edge covers, and put $\kappa(L) = +\infty$ if none exists.

The rainbow condition is necessary because choosing two different fibers of the same color would impose incompatible residues modulo the same prime.

Proposition 2.3. *Let $2 \leq i < j \leq L$ and let $q > L$ be prime. Then*

$$\begin{aligned} i! \equiv j! \pmod{q} &\iff \prod_{r=i+1}^j r \equiv 1 \pmod{q} \\ &\iff q \mid \left(\prod_{r=i+1}^j (r-1) \right). \end{aligned}$$

Proof. We have

$$j! = i! \prod_{r=i+1}^j r.$$

Also, $q > L \geq i$, so $q \nmid i!$. We may therefore cancel $i!$ modulo q . The three conditions are now equivalent. $\square$

Assigning a different prime to each index gives the following bound.

Proposition 2.4. *For every sufficiently large L ,*

$$\kappa(L) \leq (L-1) \log(20L \log L) = (1 + o(1))L \log L.$$

In particular, a rainbow edge cover exists, and the infimum in Definition 2.2 is a minimum.

Proof. The prime number theorem gives

$$\pi(20L \log L) - \pi(L) > 2L$$

for all sufficiently large L . Choose distinct primes $q_k \in (L, 20L \log L]$ for $k \in V_L$. We also have $20L \log L < L!$ for large L . The edges $e_{q_k, k!}$ form a rainbow edge cover, and their total weight is at most

$$(L-1) \log(20L \log L).$$

There are only finitely many primes between L and $L!$, and each has only finitely many fibers. Hence there are only finitely many rainbow edge covers. The nonempty finite set of their weights has a minimum. $\square$

We now relate the weight of a cover to its modulus. For sufficiently large L , let $M_{\min}(L)$ be the least squarefree product M of primes $L < q < L!$ for which there exists $A \in (\mathbb{Z}/M\mathbb{Z})^\times$ satisfying both of the following conditions:

- (1) for every prime $q \mid M$, there is a $k \in V_L$ such that $A \equiv k! \pmod{q}$;
- (2) for every $k \in V_L$, there is a prime $q \mid M$ such that $A \equiv k! \pmod{q}$.

Proposition 2.5 (Covering congruences). *For every sufficiently large L ,*

$$\kappa(L) = \log M_{\min}(L).$$

More precisely, a rainbow edge cover $\mathcal{C}$ gives a modulus

$$M(\mathcal{C}) = \prod_{q \in Q} q$$

with

$$\log M(\mathcal{C}) = w(\mathcal{C}),$$

and every pair (M, A) in the definition of $M_{\min}(L)$ gives a rainbow edge cover of weight $\log M$.

Proof. Let $\mathcal{C} = \{e_{q,a_q} : q \in Q\}$ be a rainbow edge cover. The primes in Q are distinct, so the Chinese remainder theorem gives a unique residue class $A \pmod{M(\mathcal{C})}$ satisfying

$$A \equiv a_q \pmod{q} \quad (q \in Q).$$

Every a_q is nonzero, so $(A, M(\mathcal{C})) = 1$. If $k \in E_{q,a_q}$, then $A \equiv k! \pmod{q}$. Thus $(M(\mathcal{C}), A)$ satisfies the defining conditions for $M_{\min}(L)$. Also,

$$\log M(\mathcal{C}) = \sum_{q \in Q} \log q = w(\mathcal{C}).$$

Conversely, let (M, A) satisfy the defining conditions for $M_{\min}(L)$. For each prime $q \mid M$, choose the edge $e_{q,A \bmod q}$. The residue is nonzero because $(A, M) = 1$, and the second condition says that these edges cover V_L . They use at most one edge of each color and have total weight

$$\sum_{q \mid M} \log q = \log M.$$

Taking the two minima proves the proposition. $\square$

2.1. Finite covers and a lower bound. Let $m \geq 1$, let $\mathbf{h} = (h_1, \dots, h_m)$ be distinct nonzero integers, and let $\mathcal{P}$ be a finite set of primes, none dividing $\prod_i h_i$. For $q \in \mathcal{P}$, use the nonempty fibers

$$E_{q,a}(\mathbf{h}) = \{i : h_i \equiv a \pmod{q}\}$$

with weight $\log q$. Define $\kappa_{\mathcal{P}}(\mathbf{h})$ by the same definition, with value $+\infty$ if no cover exists. The proof of Proposition 2.5 applies here as well. Choosing at most one nonzero residue for each prime gives a compatible system, and its weight is $\log M$. The minimum depends on the specified prime set $\mathcal{P}$.

Equivalently, the minimum is given by the integer program

$$\min \sum_{q,a} x_{q,a} \log q, \quad \sum_{q,a:i \in E_{q,a}} x_{q,a} \geq 1, \quad \sum_a x_{q,a} \leq 1, \quad x_{q,a} \in \{0, 1\}.$$

The sums run over nonempty fibers. Allowing real values $x_{q,a} \geq 0$ gives the corresponding linear programming relaxation. Its minimum is a lower bound for the minimum above.

Proposition 2.6 (A lower bound for the cover weight). *Let $y_1, \dots, y_m$ be nonnegative real numbers, and set*

$$z_q = \max \left\{ 0, \max_a \sum_{i \in E_{q,a}(\mathbf{h})} y_i - \log q \right\}.$$

Then

$$\kappa_{\mathcal{P}}(\mathbf{h}) \geq \sum_{i=1}^m y_i - \sum_{q \in \mathcal{P}} z_q.$$

The same inequality holds for every feasible fractional solution above.

Proof. For a feasible vector x , nonnegativity, the vertex constraints, and then the color constraints give

$$\begin{aligned} \sum_i y_i &\leq \sum_{q,a} x_{q,a} \sum_{i \in E_{q,a}} y_i \\ &\leq \sum_{q,a} x_{q,a} \log q + \sum_q z_q \sum_a x_{q,a} \\ &\leq \sum_{q,a} x_{q,a} \log q + \sum_q z_q. \end{aligned}$$

Rearranging proves the assertion for both integer and fractional covers. $\square$

Proposition 2.7 (Consecutive integer shifts). *For $\mathbf{h} = (1, 2, \dots, m)$ and an admissible set $\mathcal{P}$ of at least m primes all exceeding m , let $q_1 < \dots < q_m$ be its m smallest members. Then*

$$\kappa_{\mathcal{P}}(\mathbf{h}) = \sum_{j=1}^m \log q_j.$$

If the set $\mathcal{P}$ contains the first m primes exceeding m , then this minimum is asymptotic to $m \log m$.

Proof. Reduction modulo $q > m$ is injective on $\{1, \dots, m\}$. Every fiber is a singleton, so at least m different primes are necessary, and the m smallest suffice. In the last assertion the lower bound is $m \log m$. The prime number theorem places the m th prime exceeding m below $2m \log m$ for sufficiently large m , giving the matching upper bound $m \log(2m \log m) = (1 + o(1))m \log m$. $\square$

2.2. Choosing compatible edges. To choose edges with distinct primes, let $\mathcal{F}$ be a finite family of triples

$$F = (q, a, B_F), \quad \emptyset \neq B_F \subseteq E_{q,a}.$$

The set B_F is the *designated subset* of F . Define the auxiliary conflict graph $G_{\mathcal{F}}$ on vertex set $\mathcal{F}$ by joining distinct triples $F = (q, a, B_F)$ and $F' = (q', a', B_{F'})$ whenever $q = q'$ or

$$B_F \cap B_{F'} \neq \emptyset.$$

Proposition 2.8 (An independent set of large weight). *Suppose that the conflict graph of $\mathcal{F}$ has maximum degree at most D . If $w(F) \geq 0$ for every $F \in \mathcal{F}$, then there is a subfamily $\mathcal{J} \subseteq \mathcal{F}$ with distinct prime colors and pairwise disjoint designated subsets such that*

$$\sum_{F \in \mathcal{J}} w(F) \geq \frac{1}{D+1} \sum_{F \in \mathcal{F}} w(F).$$

In the special case $w(F) = 1$ for every F , one may choose $\mathcal{J}$ so that

$$|\mathcal{J}| \geq \frac{|\mathcal{F}|}{D+1}.$$

Proof. A finite graph of maximum degree at most D has a proper coloring with at most $D + 1$ colors: order the vertices arbitrarily and color them greedily. The color classes are independent sets and partition $\mathcal{F}$. One of them has weight at least the average of their weights. Taking this color class as $\mathcal{J}$ proves the weighted inequality. Taking unit weights gives the last assertion. Independence gives distinct prime colors and pairwise disjoint designated subsets by the definition of the conflict graph. $\square$

Covering the indices in B_F individually has leading weight $|B_F| \log L$. A single prime saves a fixed proportion of this weight only when $q \leq L^{|B_F|-\varepsilon}$ for some $\varepsilon > 0$. In the quadratic construction $|B_F| = 2$, so we seek primes in the range $q \ll L^{2-\alpha}$.

Proposition 2.9 (Completing a cover). *In the finite shift system above, suppose a candidate family $\mathcal{F}$ has designated nonempty sets $B_F \subseteq E_{q_F, a_F}$, and its conflict graph has maximum degree at most Δ . Suppose there are at least m additional admissible primes, all at most T , none occurring as a candidate color. Define*

$$\sigma(F) = (|B_F| \log T - \log q_F)_+, \quad x_+ = \max\{x, 0\}.$$

Then the enlarged prime set admits a rainbow cover of weight at most

$$m \log T - \frac{1}{\Delta + 1} \sum_{F \in \mathcal{F}} \sigma(F).$$

Proof. Discard candidates with zero saving and apply Proposition 2.8 with weights $\sigma(F)$. The selected family $\mathcal{J}$ has disjoint designated sets and saves at least $(\Delta + 1)^{-1} \sum_F \sigma(F)$. Assign a different additional prime to each remaining index. If $v = \sum_{F \in \mathcal{J}} |B_F|$, the total weight is at most

$$\sum_{F \in \mathcal{J}} \log q_F + (m - v) \log T = m \log T - \sum_{F \in \mathcal{J}} \sigma(F).$$

All added primes are different from the selected colors, so the cover is compatible. $\square$

3. PRIME DIVISORS OF A QUADRATIC

We first count the indices for which $i^2 + 3i + 1$ has a sufficiently large prime divisor. Throughout this section, put

$$f(t) = t^2 + 3t + 1.$$

3.1. Roots modulo prime powers. We also let χ_5 be the primitive quadratic character modulo 5, and we extend it by setting $\chi_5(5) = 0$. We will use the prime number theorem, the Siegel–Walfisz theorem for the fixed modulus 5, and Mertens’s estimates in the forms stated in [7, Theorems 3.4, 8.1 and 12.1]. Let

$$\Theta_{\chi_5}(x) = \sum_{p \leq x} \chi_5(p) \log p.$$

For $a \in (\mathbb{Z}/5\mathbb{Z})^\times$, the fixed-modulus Siegel–Walfisz estimate for $\pi(x; 5, a)$ and partial summation give $\vartheta(x; 5, a) = x/4 + O(xe^{-c\sqrt{\log x}})$ for some $c > 0$. Since the sum of $\chi_5(a)$ over these four classes is zero, summing with weights $\chi_5(a)$ gives $\Theta_{\chi_5}(x) \ll xe^{-c\sqrt{\log x}}$. Partial summation therefore gives

$$(2) \quad \sum_{p \leq x} \chi_5(p) \frac{\log p}{p-1} = \frac{\Theta_{\chi_5}(x)}{x-1} + \int_2^x \frac{\Theta_{\chi_5}(t)}{(t-1)^2} dt + O(1) \\ = O(1).$$

Indeed, the integrand is $O(e^{-c\sqrt{\log t}}/t)$, which is integrable on $[2, \infty)$. The same argument, now using the weight $1/(t \log t)$, gives

$$\sum_{p \leq x} \frac{\chi_5(p)}{p} = \frac{\Theta_{\chi_5}(x)}{x \log x} + \int_2^x \Theta_{\chi_5}(t) \frac{\log t + 1}{t^2 (\log t)^2} dt + O(1) = O(1).$$

For $p \neq 5$, we have $\mathbf{1}_{\chi_5(p)=1} = (1 + \chi_5(p))/2$. Mertens’s estimate now gives

$$(3) \quad \sum_{\substack{p \leq x \\ \chi_5(p)=1}} \frac{1}{p} = \frac{1}{2} \log \log x + O(1).$$

The identity

$$(4) \quad (2t+3)^2 = 4f(t) + 5$$

shows that, for $p \neq 2, 5$, the number of roots of f modulo p is $1 + \left(\frac{5}{p}\right)$. Since $5 \equiv 1 \pmod{4}$, quadratic reciprocity gives

$$\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right) = \chi_5(p).$$

Thus,

$$(5) \quad \rho(p) = 1 + \chi_5(p) \in \{0, 2\}.$$

The derivative is $f'(t) = 2t + 3$. If a root of f modulo p were not simple, then $f(t) \equiv f'(t) \equiv 0 \pmod{p}$, and (4) would give $p \mid 5$. Hence, every root under consideration is simple. Hensel’s lemma then gives exactly one lift of each root to every modulus p^a . There is no root modulo 2. Modulo 5 there is one root, but it does not lift to a root modulo 25, because

$$(6) \quad f(1+5t) = 5(1+5t+5t^2).$$

3.2. Large prime factors.

Lemma 3.1. *For all sufficiently large N ,*

$$\# \left\{ 1 \leq i \leq N : P^+(f(i)) > \frac{1}{8} i \log i \right\} \geq \frac{N}{5}.$$

Proof. We begin with the indices for which every prime factor of $f(i)$ is at most i . Put

$$E_N = \left\{ i : \frac{N}{\log N} < i \leq N, P^+(f(i)) \leq i \right\}, \quad Q_N = \prod_{i \in E_N} f(i).$$

Every prime divisor of Q_N is at most N . For $p \neq 2, 5$, the root count modulo prime powers gives

$$\text{ord}_p \left(\prod_{i \leq N} f(i) \right) = \sum_{a \geq 1} \#\{i \leq N : p^a \mid f(i)\}.$$

For every $a \geq 1$, there are $\rho(p)$ root classes modulo p^a , and hence

$$\#\{i \leq N : p^a \mid f(i)\} \leq \frac{N\rho(p)}{p^a} + \rho(p).$$

Also, $p^a \mid f(i)$ is possible only when $p^a \leq N^2 + 3N + 1$. Consequently, only $O(\log N / \log p)$ values of a occur, and summing the geometric series gives

$$\text{ord}_p \left(\prod_{i \leq N} f(i) \right) \leq \frac{N\rho(p)}{p-1} + O\left(\rho(p) \frac{\log N}{\log p}\right).$$

The prime 2 contributes nothing. Equation (6) shows that $5 \mid f(i)$ precisely when $i \equiv 1 \pmod{5}$, and that in this case $5^2 \nmid f(i)$. Thus, the contribution of $p = 5$ to the logarithm below is $O(N)$. Moreover,

$$\log N \sum_{p \leq N} \rho(p) \ll \log N \pi(N) \ll N.$$

Therefore, Mertens's estimate together with (2) gives

$$\begin{aligned} \log Q_N &\leq N \sum_{p \leq N} \frac{(1 + \chi_5(p)) \log p}{p-1} + O\left(\log N \sum_{p \leq N} \rho(p)\right) + O(N) \\ &= N \log N + O(N). \end{aligned}$$

For the corresponding lower bound, uniformly for $N/\log N < i \leq N$, we have

$$\log f(i) = 2 \log N + O(\log \log N).$$

It follows that

$$\log Q_N \geq |E_N| (2 \log N - O(\log \log N)).$$

Comparing this inequality with the upper estimate for $\log Q_N$, we obtain

$$|E_N| \leq \left(\frac{1}{2} + o(1)\right) N.$$

The interval $(N/\log N, N]$ contains $N - o(N)$ integers. Thus, at least $(1/2 - o(1))N$ integers in this interval satisfy $P^+(f(i)) > i$. Put

$$\mathcal{S}_N = \left\{ i : \frac{N}{\log N} < i \leq N, P^+(f(i)) > i \right\}.$$

We have proved that $|\mathcal{S}_N| \geq (1/2 - o(1))N$. It remains to strengthen the lower bound for these prime factors.

Discard from $\mathcal{S}_N$ the indices for which

$$P^+(f(i)) \leq \frac{1}{8}N \log N.$$

For every discarded index, the prime $q = P^+(f(i))$ exceeds i . Hence, the relevant indices lie in $[1, q - 1]$ and are distinct modulo q . Since f has at most two roots modulo q , a fixed prime q can occur for at most two indices. Therefore, the number of discarded indices is at most

$$2\pi \left(\frac{1}{8}N \log N \right) = \left(\frac{1}{4} + o(1) \right) N.$$

At least $(1/2 - o(1))N$ indices were available, and at most $(1/4 + o(1))N$ were discarded. Thus at least $(1/4 - o(1))N$ remain. For each remaining index,

$$P^+(f(i)) > \frac{1}{8}N \log N \geq \frac{1}{8}i \log i.$$

Since $1/4 - o(1) > 1/5$ for all sufficiently large N , at least $N/5$ indices satisfy the required inequality. This proves the lemma. $\square$

4. A SIEVE ESTIMATE FOR THE COMPLEMENTARY FACTOR

A prime used to cover a pair must also satisfy the upper bound needed for the modulus. Write $f(i) = mq$, where $q > L$ is prime. A lower bound for m gives an upper bound for q . We therefore estimate the number of indices for which m is small.

4.1. The polynomials to be sieved. For $0 < \alpha \leq 1/2$, let

$$E_\alpha(L) = \# \{ 1 \leq i \leq L : f(i) = mq \text{ for some } m \leq L^\alpha \text{ and prime } q > L \}.$$

Thus, $E_\alpha(L)$ counts the indices for which a prime factor $q > L$ has a complementary factor $m \leq L^\alpha$. The next lemma bounds this number in terms of α .

Lemma 4.1. *There is an absolute constant $C_0 \geq 1$ such that, uniformly for $0 < \alpha \leq 1/2$ and all sufficiently large L ,*

$$E_\alpha(L) \leq C_0 \left(\alpha L + \frac{L}{\log L} + L^{(1+\alpha)/2} (\log L)^2 \right).$$

The lower threshold for L may be chosen independently of α .

Proof. We begin by counting the roots of f modulo m . For $m \geq 1$, write

$$R(m) = \#\{a \pmod{m} : f(a) \equiv 0 \pmod{m}\}.$$

The prime-power values from the previous section are

$$R(2^e) = 0 \quad (e \geq 1), \quad R(5) = 1, \quad R(5^e) = 0 \quad (e \geq 2),$$

and, for $p \neq 2, 5$,

$$R(p^e) = 1 + \chi_5(p) \quad (e \geq 1).$$

The Chinese remainder theorem therefore gives

$$(7) \quad R(m) = \begin{cases} 2^{\omega(m/(m,5))}, & 2 \nmid m, 25 \nmid m, \text{ and} \\ & \chi_5(p) = 1 \text{ for every } p \mid m, p \neq 5, \\ 0, & \text{otherwise.} \end{cases}$$

Now fix $m \leq L^\alpha$ and a root $a \pmod{m}$, represented by $0 \leq a < m$. Every integer $i \equiv a \pmod{m}$ can be written as $i = a + mt$. Accordingly, let

$$i = a + mt, \quad T_{a,m} = \{t \in \mathbb{Z} : 1 \leq a + mt \leq L\}, \quad H = |T_{a,m}|.$$

The length of this interval of t values is

$$H = \frac{L}{m} + O(1), \quad H \asymp \frac{L}{m}, \quad H \geq L^{1/2} - 1.$$

Since $m \mid f(a)$, we can divide $f(a + mt)$ by m and obtain the integral polynomial

$$G_{a,m}(t) = \frac{f(a + mt)}{m} = mt^2 + (2a + 3)t + \frac{f(a)}{m}.$$

This polynomial has discriminant 5. It is also primitive. Indeed, if a prime divided all three coefficients, then its square would divide $(2a+3)^2 - 4f(a) = 5$, which is impossible.

If $\ell \nmid 10m$ is prime, then the affine map $t \mapsto a + mt$ is a bijection modulo ℓ . Consequently,

$$\#\{t \pmod{\ell} : G_{a,m}(t) \equiv 0 \pmod{\ell}\} = 1 + \chi_5(\ell).$$

Let

$$\mathcal{P}_0 = \{\ell : \ell \text{ is prime and } \ell \nmid 10\}, \quad \mathcal{P}_m = \{\ell \in \mathcal{P}_0 : \ell \nmid m\},$$

and extend ρ multiplicatively to squarefree integers supported on $\mathcal{P}_m$, with $\rho(d) = 0$ otherwise. For supported squarefree d , counting the root classes inside $T_{a,m}$ gives

$$(8) \quad \#\{t \in T_{a,m} : d \mid G_{a,m}(t)\} = H \frac{\rho(d)}{d} + r_d, \quad |r_d| \leq \rho(d).$$

4.2. Application of the fundamental lemma. We apply the fundamental lemma with constants independent of m and a . Define the nonnegative weights

$$a_n = \#\{t \in T_{a,m} : G_{a,m}(t) = n\}, \quad X_{a,m} = \sum_n a_n = H.$$

Thus, repeated values of $G_{a,m}$ are counted with their multiplicities. For d supported on $\mathcal{P}_m$, equation (8) says

$$\sum_{\substack{n \geq 1 \\ d|n}} a_n = H \frac{\rho(d)}{d} + r_d.$$

This is precisely Axiom 1 of [7, pp. 185–186], with local density $\nu(d) = \rho(d)$ and remainder r_d . We also have $0 \leq \rho(\ell) < \ell$ for every $\ell \in \mathcal{P}_m$.

To check the dimension condition, Mertens's estimate and (2) give, uniformly for $w \geq 2$,

$$(9) \quad \sum_{\substack{\ell \leq w \\ \ell \nmid 10}} \frac{\rho(\ell) \log \ell}{\ell} = \sum_{\ell \leq w} \frac{\log \ell}{\ell} + \sum_{\ell \leq w} \chi_5(\ell) \frac{\log \ell}{\ell} + O(1) = \log w + O(1).$$

Thus, every finite truncation of the base prime set $\mathcal{P}_0$ satisfies Axiom 2' of [7, p. 188] with $\kappa = 1$, uniformly in the truncation point. The additional local condition in that axiom holds, for example, with $k = 2$ and $\varepsilon = 1/2$, since $\rho(\ell) \in \{0, 2\}$ and every prime in $\mathcal{P}_0$ for which $\rho(\ell) = 2$ is at least 11. The cited source therefore gives Axiom 2 for every finite truncation of $\mathcal{P}_0$, with the same absolute constant C .

To obtain Axiom 2 for $\mathcal{P}_m$, observe that deleting primes can only decrease its defining product. If $3/2 \leq y_1 \leq y_2$, then

$$\prod_{\ell \in \mathcal{P}_m \cap (y_1, y_2]} \left(1 - \frac{\rho(\ell)}{\ell}\right)^{-1} \leq \prod_{\ell \in \mathcal{P}_0 \cap (y_1, y_2]} \left(1 - \frac{\rho(\ell)}{\ell}\right)^{-1}.$$

The same inequality holds for each finite truncation. Axiom 2 for $\mathcal{P}_0$ therefore gives Axiom 2 for $\mathcal{P}_m$, with the same absolute constants. In particular, these constants are independent of m and a .

Set $z = H^{1/20}$ and

$$\mathcal{Q}_{m,z} = \{\ell \in \mathcal{P}_m : \ell \leq z\}, \quad P_m(z) = \prod_{\ell \in \mathcal{Q}_{m,z}} \ell.$$

For sufficiently large L , we have $z < L$. Therefore, every prime value $G_{a,m}(t) > L$ is counted by the sifted set $\{t \in T_{a,m} : (G_{a,m}(t), P_m(z)) = 1\}$. Let $S_{a,m}$ denote the number of $t \in T_{a,m}$ for which $G_{a,m}(t)$ is a prime exceeding L . If $\mathcal{Q}_{m,z}$ is empty, the estimate below is immediate. Otherwise, let y be its largest prime. We apply [7, Theorem 18.11(a), p. 190] to the finite prime set $\mathcal{Q}_{m,z}$ with $u = 10$. In the notation of that theorem, its main term is

$$\left(1 + O_{\kappa,C}(u^{-u/2})\right) HV_m(z),$$

and its remainder is bounded by

$$O \left(\sum_{\substack{d \leq y^u \\ d|P_m(z)}} |r_d| \right).$$

Since $y \leq z$, we have $y^u \leq z^{10} = H^{1/2}$. Taking an upper bound and enlarging the remainder sum, we obtain

$$(10) \quad S_{a,m} \ll HV_m(z) + \sum_{\substack{d \leq H^{1/2} \\ d|P_m(z)}} |r_d| \leq HV_m(z) + \sum_{d \leq H^{1/2}} \rho(d),$$

where

$$V_m(z) = \prod_{\substack{\ell \leq z \\ \ell \nmid 10m}} \left(1 - \frac{\rho(\ell)}{\ell} \right).$$

For the remainder, $\rho(d) \leq \tau(d)$ gives

$$\sum_{d \leq H^{1/2}} \rho(d) \leq \sum_{ab \leq H^{1/2}} 1 \ll H^{1/2} \log H.$$

It remains to estimate the sieve product. First put

$$V_0(z) = \prod_{\substack{\ell \leq z \\ \ell \nmid 10}} \left(1 - \frac{\rho(\ell)}{\ell} \right).$$

By (3) and the convergence of $\sum_p p^{-2}$, we have

$$\log V_0(z) = \sum_{\substack{\ell \leq z \\ \chi_5(\ell)=1}} \log \left(1 - \frac{2}{\ell} \right) = -2 \sum_{\substack{\ell \leq z \\ \chi_5(\ell)=1}} \frac{1}{\ell} + O(1) = -\log \log z + O(1).$$

To account for the primes dividing m , define

$$h(m) = \prod_{\substack{p|m, p \neq 5 \\ \chi_5(p)=1}} \left(1 - \frac{2}{p} \right)^{-1}.$$

Then, omitting from $V_0(z)$ the primes that divide m gives

$$V_m(z) = V_0(z) \prod_{\substack{\ell|m, \ell \leq z \\ \ell \neq 5, \chi_5(\ell)=1}} \left(1 - \frac{2}{\ell} \right)^{-1} \ll \frac{h(m)}{\log z} \ll \frac{h(m)}{\log L}$$

uniformly for $m \leq L^\alpha$. The last comparison follows from $H \geq L^{1/2} - 1$, which gives $\log z \asymp \log L$. Substituting this into (10), we obtain

$$(11) \quad S_{a,m} \ll \frac{Hh(m)}{\log L} + H^{1/2} \log L.$$

4.3. Summation over complementary factors. If i is counted by $E_\alpha(L)$, then for at least one representation $f(i) = mq$, the residue class $a \equiv i \pmod{m}$ is one of the $R(m)$ roots and $G_{a,m}(t) = q$ for the corresponding t . Summing over m and its root classes gives an upper bound, since an index may have more than one such representation. Since $m \leq L^\alpha \leq L^{1/2}$, we also have

$$H \leq \frac{L}{m} + 1 \leq \frac{2L}{m}.$$

Put $Y := L^\alpha$. Equation (11) now gives

$$(12) \quad E_\alpha(L) \ll \frac{L}{\log L} \sum_{m \leq Y} \frac{R(m)h(m)}{m} + L^{1/2} \log L \sum_{m \leq Y} \frac{R(m)}{\sqrt{m}}.$$

The two sums on the right are estimated separately. For the first, define $g(m) = R(m)h(m)$. Equation (7) and the definition of h show that g is nonnegative and multiplicative, and that it vanishes outside the admissible support. If $p \neq 5$ and $\chi_5(p) = 1$, then

$$1 + \sum_{e \geq 1} \frac{g(p^e)}{p^e} = 1 + \frac{2p}{(p-1)(p-2)} = 1 + \frac{2}{p} + O(p^{-2}).$$

The prime 5 contributes the bounded local factor $1 + 1/5$, while inert primes and 2 contribute 1. Therefore, positivity, the Euler product, and (3) give, for $Y \geq 1$,

$$\begin{aligned} \sum_{m \leq Y} \frac{R(m)h(m)}{m} &\leq \prod_{p \leq Y} \left(1 + \sum_{e \geq 1} \frac{g(p^e)}{p^e} \right) \\ &\ll \exp \left(2 \sum_{\substack{p \leq Y \\ \chi_5(p)=1}} \frac{1}{p} + O(1) \right) \ll \log(2Y). \end{aligned}$$

For the second sum, we use $R(m) \leq \tau(m)$. Thus,

$$\begin{aligned} \sum_{m \leq Y} \frac{R(m)}{\sqrt{m}} &\leq \sum_{ab \leq Y} \frac{1}{\sqrt{ab}} \\ &\ll \sum_{a \leq Y} \frac{1}{\sqrt{a}} \sqrt{\frac{Y}{a}} \ll \sqrt{Y} \log(2Y). \end{aligned}$$

Finally,

$$\log(2Y) \leq \alpha \log L + O(1),$$

uniformly for $0 < \alpha \leq 1/2$. Substitution of these bounds into (12) gives

$$E_\alpha(L) \ll \alpha L + \frac{L}{\log L} + L^{(1+\alpha)/2} (\log L)^2,$$

with an absolute implied constant and an L -threshold independent of α . This proves the lemma. $\square$

5. COVERS FROM QUADRATIC CONGRUENCES

We now combine Lemmas 3.1 and 4.1. We choose disjoint pairs with distinct primes and assign individual primes to the remaining indices.

Proposition 5.1. *Let C_0 be as in Lemma 4.1, and set*

$$\alpha_0 = \frac{1}{1000C_0}, \quad \delta_0 = \frac{\alpha_0}{200}, \quad \eta_0 = \frac{\delta_0}{2}.$$

For every sufficiently large L , there exist a set $I \subseteq \{2, \dots, L-2\}$, distinct primes q_i for $i \in I$, and distinct primes r_k , indexed by

$$U = \{2, \dots, L\} \setminus \bigcup_{i \in I} \{i, i+2\}.$$

Put

$$M = \prod_{i \in I} q_i \prod_{k \in U} r_k.$$

These objects satisfy the following conditions:

- (1) *The pairs $\{i, i+2\}$, $i \in I$, are pairwise disjoint, and $|I| \geq L/100$.*
- (2) *For every $i \in I$,*

$$L < q_i < 2L^{2-\alpha_0}, \quad q_i \mid f(i).$$

- (3) *For every $k \in U$,*

$$L < r_k \leq 20L \log L.$$

- (4) *The primes q_i , $i \in I$, and r_k , $k \in U$, are pairwise distinct.*
- (5) *The modulus M satisfies*

$$\log M \leq (1 - \delta_0)L \log L, \quad M \leq ((L+1)!)^{1-\eta_0}.$$

Moreover, every prime divisor of M is less than $L!$.

Proof. The choice of α_0 in the proposition and Lemma 4.1 give

$$C_0 \alpha_0 L = \frac{L}{1000}.$$

Since $\alpha_0 < 1$ is fixed, we also have

$$C_0 \left(\frac{L}{\log L} + L^{(1+\alpha_0)/2} (\log L)^2 \right) = o(L).$$

Consequently,

$$E_{\alpha_0}(L) \leq \frac{L}{500}$$

for sufficiently large L . On the other hand, Lemma 3.1 supplies at least $L/5$ indices $i \leq L$ for which

$$q_i := P^+(f(i)) > \frac{1}{8} i \log i.$$

We discard the indices $i < L/10$, the two endpoints $L - 1, L$, and those counted by $E_{\alpha_0}(L)$. The number remaining is at least

$$\frac{L}{5} - \frac{L}{10} - \frac{L}{500} - 2 = \frac{49L}{500} - 2 \geq \frac{9L}{100}$$

for sufficiently large L . For every remaining index,

$$q_i > \frac{1}{8}i \log i \geq \frac{L}{80} \log \left(\frac{L}{10} \right) > L$$

once L is sufficiently large. Write $f(i) = m_i q_i$. Then $m_i > L^{\alpha_0}$, since otherwise this index would be counted by $E_{\alpha_0}(L)$. Since $f(i) < 2L^2$ for $i \leq L$, it follows that

$$q_i < 2L^{2-\alpha_0}.$$

The identity (1) now gives

$$i! \equiv (i+2)! \pmod{q_i}.$$

For each remaining index i , let

$$F_i = (q_i, i!, B_i), \quad B_i = \{i, i+2\}.$$

The identity above shows that $B_i \subseteq E_{q_i, i!}$. We apply Proposition 2.8. Two designated pairs $\{i, i+2\}$ and $\{j, j+2\}$ intersect only when $j = i$ or $j = i \pm 2$. Thus, a fixed vertex of the auxiliary conflict graph has at most two neighbors arising from an overlap of designated pairs.

A fixed prime $q > L$ occurs for at most two starting indices, because all starts are less than q and f has at most two roots modulo q . Hence a fixed vertex has at most one additional neighbor of the same prime color. The conflict graph therefore has maximum degree at most 3. Proposition 2.8, with unit weights, gives at least

$$\frac{1}{4} \cdot \frac{9L}{100} = \frac{9L}{400}$$

vertices whose associated edges have distinct colors and whose designated pairs are pairwise disjoint. For sufficiently large L , we may therefore choose a subfamily indexed by a set I with

$$|I| = \left\lceil \frac{L}{100} \right\rceil.$$

The set of remaining indices has cardinality $|U| = L - 1 - 2|I|$. The prime number theorem gives

$$\pi(20L \log L) - \pi(L) > 2L$$

for sufficiently large L . After excluding the $|I|$ primes already used for pairs, more than $2L - |I| > |U|$ primes remain. Therefore, we may assign a distinct prime $r_k \in (L, 20L \log L]$ to every $k \in U$.

It remains to estimate the product of the selected primes. Put $r = |I|$ and $s = |U| = L - 1 - 2r$. Then

$$\begin{aligned} \log M &\leq r((2 - \alpha_0) \log L + \log 2) + s(\log L + \log \log L + \log 20) \\ &= (L - 1 - \alpha_0 r) \log L + s \log \log L + O(L) \\ &\leq \left(L - 1 - \frac{\alpha_0 L}{100} \right) \log L + L \log \log L + O(L). \end{aligned}$$

For sufficiently large L , the lower order terms satisfy

$$L \log \log L + O(L) \leq \frac{\alpha_0}{200} L \log L,$$

and therefore

$$\log M \leq \left(1 - \frac{\alpha_0}{200} \right) L \log L.$$

This is the first estimate in the proposition because $\delta_0 = \alpha_0/200$. Since

$$\log((L+1)!) = L \log L - L + O(\log L),$$

we have $\log((L+1)!) \geq L \log L - 2L$ for sufficiently large L . Since $\eta_0 = \delta_0/2$, we obtain

$$\begin{aligned} (1 - \eta_0) \log((L+1)!) - \log M &\geq (1 - \eta_0)(L \log L - 2L) - (1 - \delta_0)L \log L \\ &= (\delta_0 - \eta_0)L \log L - 2(1 - \eta_0)L > 0 \end{aligned}$$

for sufficiently large L . This proves $M \leq ((L+1)!)^{1-\eta_0}$. The displayed ranges also show that every selected prime is less than $L!$ once L is sufficiently large. Choosing L above all the preceding absolute thresholds proves the proposition. $\square$

Proof of Theorem 1.1. Choose the sets and primes from Proposition 5.1. For each $i \in I$, the identity (1) and the divisibility $q_i \mid f(i)$ give

$$i! \equiv (i+2)! \pmod{q_i}.$$

Hence the edge $e_{q_i, i!}$ covers both i and $i+2$. For each $k \in U$, the edge $e_{r_k, k!}$ covers k . All the primes are distinct, so these edges form a rainbow edge cover $\mathcal{C}_L$. Proposition 5.1 gives

$$\kappa(L) \leq w(\mathcal{C}_L) = \log M \leq (1 - \delta_0)L \log L.$$

Taking $\delta = \delta_0$ proves the theorem. $\square$

5.1. Excluding a set of primes. The proof of Proposition 5.1 gives more pairs than the construction uses. We can therefore discard a prescribed set of primes and still obtain the same bound. We also record how the weight depends on the number of chosen pairs.

Theorem 5.2. *Let $\alpha_0, \delta_0, \eta_0$ be as in Proposition 5.1. There is an absolute L_2 such that, for every $L \geq L_2$ and every set $\mathcal{F}_0$ of at most $\lfloor L/100 \rfloor$ primes, the following holds. For each integer*

$$0 \leq r \leq \lfloor L/100 \rfloor, \quad s = L - 1 - 2r,$$

there are r disjoint designated pairs with distinct primes $q_i \in (L, 2L^{2-\alpha_0}) \setminus \mathcal{F}_0$ and s distinct singleton primes in $(L, 20L \log L] \setminus \mathcal{F}_0$, all different from the pair primes, giving a full cover of modulus M with

$$(13) \quad \log M \leq (L - 1 - \alpha_0 r) \log L + s \log \log L + L \log 20.$$

All selected primes are at most L^2 and less than $L!$. For $r = \lceil L/100 \rceil$ one has, uniformly in $\mathcal{F}_0$,

$$\log M \leq (1 - \delta_0) L \log L, \quad M \leq ((L + 1)!)^{1-\eta_0}.$$

Proof. The proof of Proposition 5.1, before the final subfamily is taken, provides at least $9L/400$ pairwise disjoint pairs with distinct prime colors. Deleting all colors in $\mathcal{F}_0$ removes at most $L/100$ pairs, leaving at least $L/80$. This is more than $\lceil L/100 \rceil$ for large L , so any indicated number r may be retained.

The prime number theorem gives more than $2L$ primes in $(L, 20L \log L]$ for all sufficiently large L . Excluding the forbidden primes and the r pair colors still leaves more than s choices. The logarithmic calculation is

$$\begin{aligned} \log M &\leq r((2 - \alpha_0) \log L + \log 2) + s(\log L + \log \log L + \log 20) \\ &\leq (L - 1 - \alpha_0 r) \log L + s \log \log L + L \log 20. \end{aligned}$$

For $r = \lceil L/100 \rceil$, the last two terms are absorbed exactly as in Proposition 5.1. Since $\alpha_0 > 0$ is fixed, $2L^{2-\alpha_0} \leq L^2$ and $20L \log L \leq L^2 < L!$ eventually. Every threshold depends only on the original absolute constants, not on the size of the forbidden primes or on their locations. $\square$

Corollary 5.3 (Counting the moduli). *Under the hypotheses of Theorem 5.2, put $r = \lceil L/100 \rceil$, $s = L - 1 - 2r$, and $N = \lfloor L/80 \rfloor$. For all sufficiently large L , there are at least*

$$\binom{N}{r} \binom{4L}{s} \geq 2^L$$

distinct moduli for full covers satisfying its stated power bound and avoiding $\mathcal{F}_0$. They support a probability law such that, for every integer $d \geq 1$,

$$\mathbb{P}(d \mid M) \leq (9/10)^{\omega(d)}.$$

Proof. Fix N of the surviving pair colors. The prime number theorem gives $(20 + o(1))L$ primes in $(L, 20L \log L]$, so choose a pool of $4L$ primes avoiding $\mathcal{F}_0$ and all N pair colors. Select an r -subset of the pair pool and an s -subset of the singleton pool, uniformly and independently. Every choice gives a cover after assigning its singleton primes to the remaining indices. Unique factorization recovers both selected subsets from M , proving the number of distinct moduli. Also $\binom{4L}{s} \geq (4L/s)^s \geq 4^s \geq 2^L$ eventually.

If d has a repeated prime factor or a factor outside the pools, its divisibility probability is zero. Otherwise, if it uses u pair primes and v singleton primes, the probability is

$$\frac{(r)_u}{(N)_u} \frac{(s)_v}{(4L)_v},$$

interpreted as zero if too many primes are prescribed. Here $(a)_j = a(a-1)\cdots(a-j+1)$, and $(a)_0 = 1$. Sampling without replacement bounds the ratios by $(r/N)^u$ and $(s/(4L))^v$. Eventually $r/N \leq 9/10$ and $s/(4L) \leq 1/4$, proving the assertion. $\square$

6. COUNTING COVERING CLASSES

Different assignments of indices to primes can give the same residue class. To count distinct classes, we first bound how often a factorial residue occurs. The estimates hold for every prime $q > L$.

6.1. Equal factorial residues. For $U \subseteq \{1, \dots, L\}$, put

$$m_{q,U}(a) = \#\{k \in U : k! \equiv a \pmod{q}\}, \quad \mathcal{E}_q(U) = \sum_{a \bmod q} m_{q,U}(a)^2.$$

Proposition 6.1. *For every prime $q > L$ and $a \pmod{q}$,*

$$m_{q,\{1,\dots,L\}}(a) \leq F(L) := \min_{1 \leq h \leq L} \left(\left\lceil \frac{L}{h} \right\rceil + \frac{h(h-1)}{2} \right).$$

In particular,

$$F(L) \leq \frac{3}{2}L^{2/3} + O(L^{1/3}).$$

Proof. Partition $\{1, \dots, L\}$ into $\lceil L/h \rceil$ consecutive blocks of length at most h . Within each block join consecutive occurrences of a . If m is their total number, this produces at least $m - \lceil L/h \rceil$ links. A link of length $d < h$ starting at i gives

$$\prod_{j=1}^d (i+j) - 1 \equiv 0 \pmod{q}.$$

The polynomial is monic of degree d , and the starting indices are distinct modulo q . There are therefore at most d such links, over all blocks together. Summing over d gives the finite bound. Taking $h = \lceil L^{1/3} \rceil$ gives the asymptotic estimate. $\square$

Corollary 6.2. *For every sufficiently large L ,*

$$\kappa(L) \geq \frac{L-1}{F(L)} \log(L+1) \geq (2/3 - o(1))L^{1/3} \log L.$$

Proof. A cover using m prime colors covers at most $mF(L)$ vertices. Thus $mF(L) \geq L-1$, and each color costs at least $\log(L+1)$. $\square$

Lemma 6.3. *For every $L \geq 1$, every prime $q > L$, and every $U \subseteq \{1, \dots, L\}$,*

$$\mathcal{E}_q(U) \leq 6L^{3/2}.$$

Proof. Let $h = \lceil \sqrt{L} \rceil$ and partition the interval into $b = \lceil L/h \rceil$ blocks. Write $m_{a,B}$ for the occurrences of residue a from U in block B . Cauchy–Schwarz gives

$$\mathcal{E}_q(U) \leq b \sum_{a,B} m_{a,B}^2.$$

The diagonal terms total $|U|$. For distance $1 \leq d < h$, the polynomial in the previous proof has at most d possible starts, so there are at most d equal-residue unordered pairs of this distance within the blocks. Consequently,

$$\mathcal{E}_q(U) \leq b(|U| + h(h-1)) \leq (\sqrt{L} + 1)(2L + \sqrt{L}) \leq 6L^{3/2}.$$

The estimate counts ordered pairs, including equal indices. $\square$

The $L^{3/2}$ order of Lemma 6.3 is also the $\ell = 1$ case of [3, Theorem 1], with their interval starting at 0 and containing L indices. The proof above gives the stated constant uniformly in q .

6.2. Images of permutations.

Lemma 6.4. *Let U have $s \geq 1$ elements, and let $f_j : U \rightarrow \Omega_j$ be maps to finite sets, for $1 \leq j \leq s$. Define*

$$E_j = \#\{(u, v) \in U^2 : f_j(u) = f_j(v)\}.$$

As $(u_1, \dots, u_s)$ ranges over all permutations of U , the number N of distinct vectors $(f_1(u_1), \dots, f_s(u_s))$ satisfies

$$N \geq \frac{s! s^s}{\prod_{j=1}^s E_j}.$$

Proof. For a vector z in the image, let n_z be its number of preimages, and put $p_z = n_z/s!$. If $m_j(v) = |f_j^{-1}(v)|$, then $n_z \leq \prod_j m_j(z_j)$, since allowing repeated elements only increases the number of possible preimages. For a uniform random permutation, its j th entry is uniform on U . Thus

$$\mathbb{E} m_j(f_j(u_j)) = E_j/s.$$

Concavity of the logarithm gives

$$\begin{aligned} \log(s!) &= \sum_z p_z \log(1/p_z) + \sum_z p_z \log n_z \\ &\leq \log N + \sum_j \mathbb{E} \log m_j(f_j(u_j)) \\ &\leq \log N + \sum_j \log(E_j/s). \end{aligned}$$

For the first inequality, Jensen’s inequality gives

$$\sum_z p_z \log(1/p_z) \leq \log \left(\sum_z p_z \frac{1}{p_z} \right) = \log N.$$

Rearranging and exponentiating proves the lemma. $\square$

Theorem 6.5 (Covering classes for a fixed modulus). *For all sufficiently large L , let $\mathcal{F}_0$ be any set of at most $\lfloor L/100 \rfloor$ primes and let $0 \leq r \leq \lceil L/100 \rceil$, $s = L - 1 - 2r$. There exist a modulus M satisfying (13) and a set $\mathcal{A} \subseteq (\mathbb{Z}/M\mathbb{Z})^\times$ such that every $a \in \mathcal{A}$ covers all indices $2 \leq k \leq L$ by primes dividing M , no prime in $\mathcal{F}_0$ divides M , and*

$$(14) \quad |\mathcal{A}| \geq \frac{s! s^s}{(6L^{3/2})^s} \geq \left(\frac{s^2}{6eL^{3/2}} \right)^s.$$

For each fixed $0 < \rho \leq 1/100$, taking $r = \lceil \rho L \rceil$ gives

$$M \leq X^{1-\alpha_0\rho/4}, \quad |\mathcal{A}| \geq X^{1/2-\rho-o(1)}, \quad X = (L+1)!.$$

The error term is uniform in $\mathcal{F}_0$.

Proof. Fix the pairs and the s singleton primes $p_1, \dots, p_s$ from Theorem 5.2. Let U be the indices not in the pairs. Keep the pair congruences fixed. For each permutation $(u_1, \dots, u_s)$ of U , impose

$$a \equiv u_j! \pmod{p_j} \quad (1 \leq j \leq s).$$

Each system is compatible and reduced, and covers all the remaining indices. All systems have the same modulus M . By CRT, different vectors of singleton residues give different classes modulo M . Apply Lemma 6.4 to $f_j(u) = u! \bmod p_j$ and then Lemma 6.3. The last inequality in (14) follows from $s! \geq (s/e)^s$.

For fixed ρ , equation (13) gives $\log M \leq (1 - \alpha_0\rho)L \log L + O(L \log \log L)$. Since $\log X = L \log L - L + O(\log L)$, this implies the asserted bound for M after increasing the threshold. Also $s = (1 - 2\rho)L + O(1)$, so taking logarithms in (14) gives

$$\log |\mathcal{A}| \geq (1/2 - \rho)L \log L - O_\rho(L).$$

This proves the fixed- ρ assertions. $\square$

Corollary 6.6. *Write $\alpha = \alpha_0$. For every sufficiently large L and every forbidden set as in Theorem 6.5, there are M and $\mathcal{A}$ as in that theorem such that*

$$M \leq X e^{-5L}, \quad |\mathcal{A}| \geq X^{1/2} \exp \left(-\frac{3}{\alpha} L \log \log L \right), \quad X = (L+1)!.$$

In particular, $|\mathcal{A}| \geq X^{1/2-o(1)}$.

Proof. Choose

$$r = \left\lceil \frac{L(\log \log L + 10)}{\alpha \log L} \right\rceil.$$

Since $\alpha > 0$ is fixed, eventually $r \leq \lceil L/100 \rceil$ and $s = L - 1 - 2r \geq L/2$. Equation (13) yields

$$\log M \leq L \log L - L(\log \log L + 10) + L \log \log L + L \log 20 \leq L \log L - 7L.$$

As $\log X \geq L \log L - 2L$ eventually, $M \leq X e^{-5L}$. Taking logarithms in (14) and using $s \geq L/2$ gives, with an absolute constant C ,

$$\begin{aligned} \log |\mathcal{A}| &\geq \frac{1}{2}s \log L - CL \\ &\geq \frac{1}{2}L \log L - \frac{L}{\alpha}(\log \log L + 10) - CL - \frac{3}{2} \log L \\ &\geq \frac{1}{2} \log X - \frac{3}{\alpha}L \log \log L \end{aligned}$$

for sufficiently large L . All estimates are uniform in the forbidden set. Finally $L \log \log L = o(\log X)$. $\square$

7. TRIPLES WITH SMALL SPAN AND LOWER BOUNDS FOR COVERS

The construction above groups indices in pairs. To reduce the weight further, one may try to use fibers containing three or more indices. We now bound the number of triples of equal factorial residues whose first and last indices are close together. We count a triple separately for each prime for which the equality holds.

For $L \geq 3$ and $1 \leq D \leq L$, define

$$T(L, D) = \sum_{\substack{q > L \\ q \text{ prime}}} \#\{2 \leq i < j < k \leq L : k - i \leq D, i! \equiv j! \equiv k! \pmod{q}\}.$$

The sum is finite because each triple imposes divisibility of fixed nonzero integers. Put $\mathcal{B}(D) = 0$ for $D \leq 3$, and otherwise put

$$(15) \quad \mathcal{B}(D) = \sum_{b=4}^D \left(\frac{b(b-3)}{2} \log 2 + \frac{b^3 - 13b + 12}{6} \log b \right).$$

Theorem 7.1. *For every $L \geq 3$ and $1 \leq D \leq L$,*

$$T(L, D) \leq \frac{\mathcal{B}(D)}{\log(L+1)}.$$

Moreover,

$$\mathcal{B}(D) = \frac{D^4}{24} \log D - \frac{D^4}{96} + O(D^3 \log D).$$

Thus for fixed $c > 0$,

$$T(L, \lfloor cL^{1/4} \rfloor) \leq (c^4/96 + o(1))L,$$

and $T(L, D) = o(L)$ whenever $D = o(L^{1/4})$.

7.1. Common roots and resultants. For $d \geq 1$, set

$$P_d(x) = \prod_{j=1}^d (x + j) - 1.$$

The irreducibility of these polynomials is classical; it is recorded, for example, in the proof of [6, Theorem 2.1]. We give the proof together with the resultant bound needed below.

Lemma 7.2. *The polynomials P_d are irreducible over $\mathbb{Q}$. For $1 \leq a < b$,*

$$0 < |\text{Res}(P_a, P_b)| \leq 2^a b^{a(b-a)}.$$

Proof. If $P_d = FG$ were a nontrivial factorization, Gauss's lemma would give monic integral factors, with $1 \leq m := \deg F \leq d/2$ after interchanging them. For $j = 1, \dots, d$, $F(-j)G(-j) = -1$, hence $F(-j) = \pm 1$. Thus $F^2 - 1$ has d distinct roots. If $2m < d$ this is impossible. If $2m = d$, monicity gives

$$F^2 - 1 = \prod_{j=1}^d (x + j) = P_d + 1,$$

so $P_d = F^2 - 2$. The divisibility $F \mid P_d$ would imply $F \mid 2$, again impossible.

The distinct degrees of the irreducible polynomials imply that their resultant is a nonzero integer. If $P_a(\zeta) = 0$, then $\prod_{j=1}^a (\zeta + j) = 1$, so $|\zeta + j_0| \leq 1$ for some $j_0 \leq a$. For $a < j \leq b$ this gives $|\zeta + j| \leq b$. Consequently,

$$|P_b(\zeta)| = \left| \prod_{j=a+1}^b (\zeta + j) - 1 \right| \leq 2b^{b-a}.$$

The product of these bounds over the a roots of P_a is the asserted resultant bound. $\square$

Lemma 7.3. *If $F, G \in \mathbb{Z}[x]$ are monic nonconstant polynomials with nonzero resultant, then the number of their distinct common roots modulo a prime q is at most $v_q(\text{Res}(F, G))$.*

Proof. Let $m = \deg F$, $n = \deg G$, and consider the Sylvester map $(U, V) \mapsto UF + VG$, where $\deg U < n$ and $\deg V < m$. Monicity implies that the reductions have the same degrees, so the number d of their distinct common roots is at most $\min(m, n)$. Every polynomial in the image vanishes at these roots. Evaluation at the d points imposes d independent linear conditions on polynomials of degree less than $m + n$. Thus the integer Sylvester matrix has rank deficiency at least d modulo q .

Lift invertible row operations over $\mathbb{F}_q$ to an integral matrix whose determinant is not divisible by q . At least d rows of the resulting matrix are divisible by q , so its determinant is divisible by q^d . Its determinant differs from the original resultant by a q -adic unit. Hence $q^d \mid \text{Res}(F, G)$. $\square$

Proof of Theorem 7.1. Put $a = j - i$ and $b = k - i$. By Proposition 2.3, a triple collision gives a common root i of P_a and P_b modulo q . For fixed a, b, q , the possible starts $2 \leq i \leq L - b$ are distinct modulo $q > L$. Thus Lemma 7.3 bounds their number by $v_q(R_{a,b})$, where $R_{a,b} = \text{Res}(P_a, P_b)$. Summing over $q > L$ gives

$$\begin{aligned} \log(L+1) \sum_{q>L} \#\{\text{valid starts for } a, b, q\} &\leq \sum_{q>L} v_q(R_{a,b}) \log q \\ &\leq \log |R_{a,b}| \\ &\leq a \log 2 + a(b-a) \log b. \end{aligned}$$

Adjacent factorials do not collide in the index range: cancelling $i!$ from $(i+1)! \equiv i! \pmod{q}$ would give $q \mid i$. Therefore $2 \leq a \leq b-2$ and $b \geq 4$. Summing the last bound over these gaps yields (15), since

$$\sum_{a=2}^{b-2} a = \frac{b(b-3)}{2}, \quad \sum_{a=2}^{b-2} a(b-a) = \frac{b^3 - 13b + 12}{6}.$$

Finally, comparison with the integral gives

$$\sum_{b \leq D} b^3 \log b = \frac{D^4}{4} \log D - \frac{D^4}{16} + O(D^3 \log D).$$

The remaining sums in (15) are of lower order. This proves the expansion and its consequences. For bounded D , the $o(L)$ assertion follows directly from the finite bound. $\square$

7.2. Lower bounds for covers. We call a choice of one nonempty set $B_q \subseteq E_{q,a_q}$ for each selected prime, covering the indicated vertices, a *designated cover*. These sets may overlap. A consecutive triple consists of three successive elements in the increasing enumeration of one B_q ; the elements need not be consecutive integers.

Theorem 7.4. *Let a designated cover have weight w , and let $n = |\bigcup_q B_q|$. Let $K_{>D}$ count its prime-labelled consecutive triples of span greater than D . Then*

$$K_{>D} \geq n - \frac{2w + \mathcal{B}(D)}{\log(L+1)}.$$

In particular, if all its consecutive triples have span at most D , then

$$w \geq \frac{1}{2}(n \log(L+1) - \mathcal{B}(D)).$$

Proof. Write m for the number of selected primes and $b_q = |B_q|$. There are $\sum_q (b_q - 2)_+ \geq \sum_q b_q - 2m \geq n - 2m$ consecutive triples. At most $T(L, D)$ of them have span at most D . They are distinct as prime-labelled triples, even when different colors have overlapping designated sets. Now apply Theorem 7.1 and $m \leq w/\log(L+1)$. $\square$

Corollary 7.5. *Suppose $c > 0$ and $0 \leq u < 1$ are fixed, and a sequence of designated covers covers at least $(1-u)(L-1)$ indices. If every consecutive-triple span is at most $cL^{1/4}$, then*

$$\frac{w}{\log((L+1)!)} \geq \frac{1-u}{2} - \frac{c^4}{192} - o(1).$$

For full covers this gives $1/2 - c^4/192 - o(1)$. If every span is $o(L^{1/4})$, the full-cover lower bound is $1/2 - o(1)$.

Proof. Use $D = \lfloor cL^{1/4} \rfloor$ in Theorem 7.4 and $\log((L+1)!) = L \log L - L + O(\log L)$. For a full cover, use $n = L-1$ directly. For a varying $D = o(L^{1/4})$, use $\mathcal{B}(D)/\log L = o(L)$. $\square$

For example, if a full cover has $w \leq \theta \log((L+1)!)$ for a fixed $\theta < 1/2$, then for each fixed $c > 0$ it has at least

$$(1 - 2\theta - c^4/96 - o(1))L$$

consecutive triples of span exceeding $cL^{1/4}$, counted with their primes, whenever the coefficient is positive.

Suppose a designated set is connected by a graph whose edges join indices at distance at most h . Every consecutive gap in that set is then at most h , since a larger gap would disconnect the graph. Thus its consecutive-triple spans are at most $2h$, and the preceding inequalities apply with $D = \lfloor 2h \rfloor$.

8. ROUGH INTEGERS IN PRESCRIBED PROGRESSIONS

We now count rough integers in the residue classes given by a cover. The Chinese remainder theorem and inclusion–exclusion give the following estimate, which also allows a prescribed congruence.

Proposition 8.1 (Counting rough integers). *Let $L \geq 2$, $W = \prod_{p \leq L} p$, and let $M, d \geq 1$ satisfy $(M, dW) = 1$. Suppose $(A, M) = (b, d) = 1$. Put*

$$W_d = \prod_{p \leq L, p \nmid d} p, \quad V_d(L) = \prod_{p \leq L, p \nmid d} (1 - 1/p).$$

For real $u < v$, the number $N_{A,b}(u, v)$ of integers $n \in (u, v]$ with

$$n \equiv A \pmod{M}, \quad n \equiv b \pmod{d}, \quad (n, W) = 1$$

satisfies

$$(16) \quad \left| N_{A,b}(u, v) - \frac{v-u}{Md} V_d(L) \right| \leq 2^{\omega(W_d)}.$$

The bound is uniform in every displayed parameter.

Proof. The two prescribed congruences give a unique class modulo Md . Primes dividing d cannot divide n because $(b, d) = 1$. For each $e \mid W_d$, the additional condition $e \mid n$ gives a unique class modulo Mde , since $(e, Md) = 1$. Its count in $(u, v]$ differs from $(v-u)/(Mde)$ by at most one. Inclusion–exclusion therefore gives

$$N_{A,b}(u, v) = \sum_{e \mid W_d} \mu(e) \left(\frac{v-u}{Mde} + \epsilon_e \right), \quad |\epsilon_e| \leq 1.$$

The main term factors as $\frac{v-u}{Md} \prod_{p \mid W_d} (1 - 1/p)$, and there are $2^{\omega(W_d)}$ error terms. $\square$

Let $\mathcal{A}$ be a set of distinct classes modulo a common modulus M , each giving a full cover of $2!, \dots, L!$ with primes below $L!$. Assume $(M, d) = 1$ and $(b, d) = 1$. For $X = (L+1)!$, formula (16) gives at least

$$(17) \quad |\mathcal{A}| \left(\frac{X - 2L!}{Md} V_d(L) - 2^{\omega(W_d)} \right)$$

L -rough integers in these classes and in $b \pmod{d}$. The classes are disjoint, because equality of the combined classes modulo Md implies equality modulo M .

All these integers satisfy the factorial-difference conditions. For $2 \leq k \leq L$, some covering prime satisfies

$$q \mid n - k!, \quad 1 < q < L! < n - k!.$$

Thus it is a proper divisor. Roughness makes n odd, and $n - 1 > 2$, so the case $k = 1$ follows as well.

8.1. Counting in one covering class.

Corollary 8.2. *Let G_L count the auxiliary solutions in $(2L!, X]$, $X = (L+1)!$. For every sufficiently large L ,*

$$G_L \geq (X - 2L!)e^{-\kappa(L)} \prod_{p \leq L} (1 - 1/p) - 2^{\pi(L)}.$$

More generally, for any fixed $\lambda > \log 2$, uniformly over full covers with

$$M \leq X \exp\left(-\lambda \frac{L}{\log L}\right),$$

the number of L -rough integers in their single CRT class and in $(2L!, X]$ is

$$(e^{-\gamma} + o(1)) \frac{X - 2L!}{M \log L},$$

where γ is Euler's constant. Every integer counted is an auxiliary solution. This is an asymptotic for the selected progression, not for all auxiliary solutions.

Proof. Take $d = 1$ and a minimum-cost cover in Proposition 8.1 for the first assertion. For the second, Mertens's product estimate gives $V_1(L) \sim e^{-\gamma}/\log L$, and the prime number theorem gives $2^{\pi(L)} = \exp((\log 2 + o(1))L/\log L)$. The main term in (16) divided by its error bound tends to infinity uniformly under the stated modulus condition, because $\lambda > \log 2$. This proves the asymptotic. The classical estimates are [7, Theorems 3.4(c) and 8.1]. $\square$

Theorem 8.3. *Fix $0 < \eta < 1$. For every sufficiently large L , depending only on η , a full cover with $M \leq X^{1-\eta}$, $X = (L+1)!$, yields at least $X^\eta/(100 \log L)$ auxiliary solutions in $(2L!, X]$.*

Proof. Use (16) with $d = 1$ and the cover's CRT class. For sufficiently large L , $X - 2L! \geq X/2$ and $V_1(L) \geq 1/(4 \log L)$. Moreover,

$$2^{\pi(L)} = \exp(O(L/\log L)) = o\left(\frac{X^\eta}{\log L}\right).$$

The count is therefore at least $X/(16M \log L)$ for sufficiently large L , which exceeds the asserted bound. The argument preceding Corollary 8.2 shows that all the required differences are composite. $\square$

Proof of Theorem 1.2. Decrease the constant δ in Theorem 1.1 if necessary so that $0 < \delta < 1$, and put $\eta = \delta/2$. The minimum-cost cover exists for sufficiently large L by Proposition 2.4. Since

$$\log M = \kappa(L) \leq (1 - \delta)L \log L, \quad \log X = L \log L - L + O(\log L),$$

one has $M \leq X^{1-\eta}$ eventually. Theorem 8.3 proves the assertion. $\square$

8.2. Counting in a prescribed residue class.

Proof of Theorem 1.3. Write $\alpha = \alpha_0$. For a given $d \leq e^L$, let $\mathcal{F}_0$ be the set of prime divisors of d exceeding L . Then

$$|\mathcal{F}_0| \log(L+1) \leq \log d \leq L,$$

so $|\mathcal{F}_0| \leq \lfloor L/100 \rfloor$ for all sufficiently large L , uniformly in d . Use Corollary 6.6 avoiding this set. Its modulus satisfies

$$(M, d) = 1, \quad M \leq X e^{-5L}, \quad |\mathcal{A}| \geq X^{1/2} \exp\left(-\frac{3}{\alpha} L \log \log L\right).$$

All selected primes exceed L , so $(M, W) = 1$ as well.

For any $(b, d) = 1$, apply (17). Since $d \leq e^L$, $V_d(L) \geq V_1(L) \geq 1/(4 \log L)$, and $2^{\omega(W_d)} \leq 2^L$, the number of rough integers contributed by each combined class is at least

$$\frac{e^{4L}}{8 \log L} - 2^L \geq 1$$

for all sufficiently large L . Hence the total is at least $|\mathcal{A}|$. The preceding argument gives compositeness of the factorial differences. Finally,

$$\frac{(3/\alpha)L \log \log L}{\log X} \rightarrow 0.$$

For each $\varepsilon > 0$, increasing the threshold makes the lower bound at least $X^{1/2-\varepsilon}$. Every threshold is independent of d and b , as required. $\square$

9. COMPUTATIONAL EXAMPLES AND FURTHER QUESTIONS

9.1. Examples of covers. The supplementary files contain full covers at three finite values of L . Each selected prime, residue, complete fiber, and designated subset is listed. The accompanying program recomputes the residues, checks primality by trial division, and verifies that the primes are distinct and the sets cover all indices. It checks the bounds below by comparing integer powers.

L	Primes	Consecutive triples	Minimum span	Bound
300	99	106	24	$M^{100} < X^{46}$
1000	296	411	39	$M^{100} < X^{39}$
3000	810	1383	30	$M^{50} < X^{17}$

Here $X = (L + 1)!$, and the triples are consecutive triples of the designated subsets. In all three certificates every prime lies in $(L, L^2]$. No listed triple has span at most $\lfloor L^{1/4} \rfloor$. As one concrete example from the $L = 300$ certificate,

$$17! \equiv 81! \equiv 87! \equiv 176! \equiv 198! \equiv 293! \equiv 70 \pmod{337}.$$

These examples give the displayed bounds for $\kappa(L)$ at the three specified values. They give no asymptotic bound. The supplementary programs also check the inequality in Lemma 6.4, the progression counts, and resultants for small degrees. These computations are separate from the proofs.

9.2. Further questions. The upper and lower bounds for the minimum weight are far apart:

$$(2/3 - o(1))L^{1/3} \log L \leq \kappa(L) \leq (1 - \delta)L \log L.$$

Our arguments do not determine its order or prove $\kappa(L) = o(L \log L)$. Corollary 7.5 gives one restriction on improving the upper bound: a cover whose weight is less than half the leading weight of the individual assignments, by a fixed proportion, must contain a positive proportion of triples with large span. Choosing such triples with distinct primes and disjoint indices is a further problem.

Proposition 2.9 gives a way to use additional equalities between factorial residues. A compatible family of designated subsets with a large total saving $\sum_F (|B_F| \log T - \log q_F)$ gives a cover of small weight, provided enough unused primes of size at most T are available for the remaining indices. The difficulty is to find such subsets and choose them without repeating primes or indices. A lower bound for the number of distinct factorial residues alone does not give this choice.

One can also ask how many covering classes a fixed modulus admits. Lemma 6.4 bounds this number using counts of equal factorial residues at each prime. Stronger estimates for these counts may improve the $X^{1/2-o(1)}$ lower bound even if the bound for $\kappa(L)$ stays the same. The integers counted by this method are rough integers; primality is a separate condition.

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USE OF AI TOOLS

The mathematical exploration and initial proof development used OpenAI's ChatGPT and Codex extensively. These tools also assisted with computation, literature searches, algebraic checks, and manuscript preparation. The

author selected and reframed the problem, proposed the broader hypergraph interpretation, and directed the successive revisions.

Aristotle (Harmonic) produced the Lean formalization of the original auxiliary theorem. The extent of that formalization is stated in the introduction.

REFERENCES

- [1] W. D. Banks, F. Luca, I. E. Shparlinski, and H. Stichtenoth, *On the value set of $n!$ modulo a prime*, Turkish J. Math. **29** (2005), no. 2, 169–174.
- [2] T. F. Bloom, *Erdős Problem 1059*, erdosproblems.com/1059, accessed August 9, 2026.
- [3] M. Z. Garaev, F. Luca, I. E. Shparlinski, *Exponential sums and congruences with factorials*, J. Reine Angew. Math. **584** (2005), 29–44.
- [4] R. K. Guy, *Unsolved Problems in Number Theory*, 3rd ed., Problem Books in Mathematics, Springer, New York, 2004, Problem A2, pp. 10–12.
- [5] X. Hu, *Factorial residues modulo a prime: beyond the square-root bound*, arXiv:2608.01781 (2026).
- [6] O. Klurman and M. Munsch, *Distribution of factorials modulo p* , J. Théor. Nombres Bordeaux **29** (2017), no. 1, 169–177.
- [7] D. Koukoulopoulos, *The Distribution of Prime Numbers*, Graduate Studies in Mathematics, vol. 203, American Mathematical Society, Providence, RI, 2019.
- [8] J. Land, *A proposed solution to Erdős Problem 1059: prime avoidance for a sparse set of shifts*, preprint, September 5, 2026, github.com/beetree/math_erdos_1059.
- [9] Palomar Registry, *Factorial-residue hypergraphs and an auxiliary problem of Erdős*, entry PALOMAR-2026-08-27-000003 v1, palomar-registry.org, 2026.